

PHILIPPINE NATIONAL STANDARD

PNS/BAFS 234:2018
ICS 67.080.20

Baby corn – Grading and classification



BUREAU OF PHILIPPINE STANDARDS (BPS)

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Foreword

The Philippine National Standard PNS/BAFS 234:2018 Baby corn – Grading and classification is a modified adoption of the CODEX STAN 188-1993 Standard for Baby Corn. With the initiative of the Bureau of Agriculture and Fisheries Standards (BAFS), a Technical Working Group (TWG) for the adoption of the Standard was created and authorized under Special Order No. 604 Series of 2017. The TWG was composed of the following agencies and institutions: National Food Authority – Food Development Center (NFA-FDC), Bureau of Animal Industry (BAI), Food and Nutrition Research Institute (FNRI), Philippine Center for Postharvest Development and Mechanization (PHilMech), University of the Philippines Los Baños (UPLB), and Department of Agriculture (DA) – DA Corn Program. This Standard intends to provide guidance in the regulation and trade and ensure consistent quality and safety of baby corn.

In this Standard, modifications due to national legal requirements and the particular needs of the Philippine industry have been made. These technical deviations have been incorporated in the provisions of the Standard. Some provisions of the Standard were also adopted from the ASEAN Standard for Baby Corn (ASEAN Stan 52:2017).

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Baby corn – Grading and classification

1 Scope

This Standard applies to baby corn grown from *Zea mays* L., of the Gramineae family to be supplied fresh to the consumer, after preparation and packaging. Baby corn for industrial processing is excluded.

2 Normative references

The titles of the standards publications referred to in this standard are listed on the inside back cover.

3 Definitions

For the purpose of this standard, the following definitions apply.

3.1**baby corn**

also known as young corn, is the ear shoot of corn without the silk and husk

3.2**sound**

parameter that indicates good quality

4 Provisions concerning quality**4.1 Minimum requirements**

4.1.1 In all classes, subject to the special provisions for each class and the tolerances allowed, the baby corn must be:

- whole;
- fresh in appearance;
- sound, not affected by rotting or deterioration such as to make it fit for consumption;
- clean, practically free of any visible foreign matter;
- free of any foreign smell and/or taste; practically free of silk;
- practically free of pests and damage caused by them;
- practically free from mechanical and/or physiological damage such as low and/or high temperature; and
- practically free of surface moisture excluding condensation.

The cut that is made on the base of the ear shoot should be clean and well defined. A slight discoloration of the cut surface due to storage is acceptable.

4.1.2 The baby corn must have been harvested and have reached an appropriate degree of development in accordance with criteria proper to the variety and the area in which they are grown. The development and condition of the baby corn must be such as to enable it:

- to withstand transport and handling; and
- to arrive in satisfactory condition at the place of destination.

4.2 Classification

The baby corn is classified in three classes defined below:

4.2.1 “Extra” class

The baby corn in this class must be well cut and trimmed, free of husk, stalk and silk, intact, and of superior quality. They must be characteristic of the variety and/or commercial type. They must be free of defects, with the exception of very slight superficial defects, provided these do not affect the general appearance of the produce, the quality, the keeping quality, and presentation in the package.

4.2.2 Class I

The baby corn in this class must be well cut and trimmed, free of husk and stalk, and of good quality. They must be characteristic of the variety and/or commercial type. The following slight defects, however, may be allowed, provided these do not affect the general appearance of the produce, the quality, the keeping quality, and presentation in the package:

- slight defects in shape, color, and texture;
- slight defects in regular arrangement of undeveloped kernels (ovules);
- slight defects on the surface due to bruising, scratches, or other mechanical damage. The total area affected shall not exceed 5% per ear shoot;
- silk attached to and/or broken from the ear shoot shall be minimal without affecting the appearance.

4.2.3 Class II

This class includes baby corn which do not qualify for inclusion in the higher classes, but satisfy the minimum requirements specified in Section 4.1 above. The following defects, however, may be allowed, provided the baby corn retain their essential characteristics as regards the quality, the keeping quality, and presentation:

- defects in shape, color, and texture;
- defects in regular arrangement of undeveloped kernels (ovules);
- defects on the surface due to bruising, scratches, or other mechanical damage. The total area affected shall not exceed 10% per ear shoot;
- silk attached to and/or broken from the ear shoot shall be minimal without affecting the appearance.

5 Provisions concerning sizing

Size is determined by the length of baby corn, in accordance with the following table:

Table 1 – Sizing of baby corn

Size code	Length, (cm)
1	> 9.0 – 12.0
2	> 7.0 – 9.0
3	4.0 – 7.0

For all sizes, the minimum diameter should not be less than 1.0 cm and the maximum diameter not more than 2.0 cm.

6 Provisions concerning tolerances

6.1 Quality tolerances

Tolerances in respect of quality and size shall be allowed in each package for produce not satisfying the requirements of the class indicated.

6.1.1 “Extra” Class

Five percent by number or weight of baby corn not satisfying the requirements of the class, but meeting those of Class I or, exceptionally, coming within the tolerances of that class.

6.1.2 Class I

Ten percent by number or weight of baby corn not satisfying the requirements of the class, but meeting those of Class II or, exceptionally, coming within the tolerances of that class.

In the case of baby corn with incompletely removed husk and stalk, only 5% by number or weight of 0.5 cm long of the husk and stalk is allowed.

6.1.3 Class II

Ten percent by number or weight of baby corn satisfying neither the requirements of the class nor the minimum requirements, with the exception of produce affected by rotting or any other deterioration rendering it unfit for consumption.

In the case of baby corn with incompletely removed husk and stalk, only 5% by number or weight of 0.5 cm long of the husk and stalk is allowed.

6.2 Size tolerances

For “Extra” Class, 5%; and for Class I or Class II, 10%; by number or weight of baby corn not satisfying the requirements as regards sizing, but falling within the class immediately above or below those indicated in Section 5.

7 Provisions concerning presentation

7.1 Uniformity

The contents of each package (or lot for produce presented in bulk) must be uniform and contain only of baby corn of the same origin, quality, and size. The visible part of the contents of the package (or lot for produce presented in bulk) must be representative of the entire contents.

7.2 Packaging

The baby corn must be packed in such a way as to protect the produce properly. The primary packaging materials to be used shall be new¹, clean, and of a quality such as to avoid causing any external or internal damage to the produce. The use of materials, particularly of paper or stamps bearing trade specifications, is allowed, provided the printing or labelling has been done with non-toxic ink or glue.

The baby corn shall be packed in each container in compliance with the Recommended International Code of Practice for Packaging and Transport of Fresh Fruits and Vegetables (CAC/RCP 44-1995) or its equivalent Philippine National Standard (PNS).

The containers shall meet the quality, hygiene, ventilation and resistance characteristics to ensure suitable handling, shipping, and preserving of the baby corn. Packages (or lot for produce presented in bulk) must be free of foreign matter and odor.

8 Marking or labelling

8.1 Consumer packages

In addition to the requirements of the Codex General Standard for the Labelling of Prepackaged Foods (CODEX STAN 1-1985) or its equivalent PNS, the following specific provisions apply:

¹ For the purposes of this Standard, this includes recycled material of food-grade quality.

If the produce is not visible from the outside, each package shall be labelled as to the name of the produce and may be labelled as to the name of the variety (optional).

8.2 Non-retail containers

Each package must bear the following particulars, in letters grouped on the same side, legibly and indelibly marked, and visible from the outside, or in the documents accompanying the shipment. For produce transported in bulk, these particulars must appear on a document accompanying the goods.

8.2.1 Identification

Name and address of exporter, packer and/or dispatcher. Identification code (optional)².

8.2.2 Nature of produce

Name of the produce if the contents are not visible from the outside. Name of the variety or commercial type (optional).

8.2.3 Origin of produce

Country of origin and, optionally, district where grown or national, regional or local place name.

8.2.4 Commercial identification

- Class;
- Size (size code);
- Net weight (optional);
- Produce name;
- Variety name (optional);
- Number of units (optional); and
- Date of packing.

8.2.5 Official inspection mark (optional)

9 Contaminants

9.1 The produce covered by this Standard shall comply with the maximum levels of the Codex General Standard for Contaminants and Toxins in Food and Feed (CODEX STAN 193-1995) or its equivalent PNS.

9.2 The produce covered by this Standard shall comply with the maximum residue limits for pesticides established by the Codex Alimentarius Commission.

² The national legislation of a number of countries requires the explicit declaration of the name and address. However, in the case where a code mark is used, the reference “packer and/or dispatcher (or equivalent abbreviations)” has to be indicated in close connection with the code mark.

10 Hygiene

10.1 It is recommended that the produce covered by the provisions of this Standard be prepared and handled in accordance with the appropriate sections of the Recommended International Code of Practice – General Principles of Food Hygiene (CAC/RCP 1-1969), Code of Hygienic Practice for Fresh Fruits and Vegetables (CAC/RCP 53-2003), and other relevant Codex texts and PNS such as Codes of Hygienic Practice and Codes of Practice.

10.2 The produce should comply with any microbiological criteria established in accordance with the Principles for the Establishment and Application of Microbiological Criteria for Foods (CAC/GL 21-1997).

References

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The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application.

CODEX STAN 1-1985 Codex General Standard for the Labelling of Prepackaged Foods

CODEX STAN 193-1995 Codex General Standard for Contaminants and Toxins in Food and Feed

CAC/RCP 1-1969 Recommended International Code of Practice – General Principles of Food Hygiene

CAC/RCP 44-1995 Code of Practice for Packaging and Transport of Fresh Fruit and Vegetables

CAC/RCP 53-2003 Code of Hygienic Practice for Fresh Fruits and Vegetables

CAC/GL 21-1997 Principles and Guidelines for the Establishment and Application of Microbiological Criteria Related to Foods



Your partner in product safety

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